Alunos: Felipe Echeverria Ubalva RGM: 45611
e Bruno henrique Tuneca Estigarribia
RGM: 43555

1. $\begin{cases} 3x - 9y + 3z = 2 \\ 3x - 5y + 9z = 2 \\ x + 3y + 3z = a \end{cases}$ $\rightarrow$ $-2 \cdot L_1 + L_2$ $\begin{cases} -6x + 18y - 9z = -4 \\ 8x - 5y + 9z = 2 \end{cases}$

$-3x + 13y = -2$

$-1 \cdot L_1 + L_3$ $\begin{cases} -3x + 9y - 3z = -2 \\ x + 3y + 3z = a \end{cases}$

$-2x + 12y = -2 + a$

$\begin{cases} -3x + 13y = -2 \\ -2x + 12y = -2 + a \end{cases}$

$-1(L_1 + 1) + L_2$ $\begin{cases} 2x - 14y = 1 \\ -2x + 12y = -2 + a \end{cases}$

$-2y = 1 + (-2 + a)$

$-2y = 1 - 2 + a$

$-2y = -1 + a$

$-y = \frac{-1 + a}{2}$ $(\cdot -1)$

$y = \frac{1 + a}{2}$

$2 \cdot (-1 + a) + 12 \cdot (1 + a) = -2$

$$\begin{cases} \\ -2x + 12y = -2 + a \end{cases}$$

$$-2y = 1 + (-2 + a)$$

$$-2y = 1 - 2 + a$$

$$-2y = -1 + a$$

$$-y = \frac{-1 + a}{2} \quad (\cdot -1)$$

$$y = \frac{1 + a}{2}$$

$$-2x + 12\left(\frac{1+a}{2}\right) = -2 + a$$

$$-2x + \frac{12 + 12a}{2} = -2 + a$$

$$-2x + 6 + 6a = -2 + a$$

$$-2x = -2 - 6 + a - 7a$$

$$-2x = -8 - 6a$$

$$-x = -8 - 6a \quad (-1)$$

$$x = \frac{8 + 6a}{2}$$

$$-3 \cdot \left(\frac{8 + 6a}{2}\right) + 13 \cdot \left(\frac{1 + a}{2}\right) = -2$$

$$\frac{-24 - 18a}{2} + \frac{13 + 13a}{2} = -2$$

$$\frac{-24 + 13 - 18 + 13}{2} = -2$$

$$\frac{-11 - 2a}{2} = -2$$

$$-11 - 2a = -2 \cdot 2$$

$$-2a = -4 + 11$$

$$-a = \frac{7}{2}$$

$$\boxed{a = \frac{-7}{2}}$$

FORONI

Scanned with CamScanner

2) utilizando o teorema de Laplace

$$A = \begin{array}{c} C_{11} \\ C_{21} \\ C_{31} \\ C_{41} \end{array} \begin{vmatrix} 1 & 4 & 2 & 4 \\ 1 & 2 & 1 & 2 \\ 1 & 2 & 0 & 0 \\ 0 & -1 & 1 & 2 \end{vmatrix}$$

$C_{ij} = (-1)^{i+j} \cdot d_{ij}$

$1 \cdot C_{11} + 1 \cdot C_{21} + 1 \cdot C_{31} + \cancel{0 \cdot C_{41}}$

$C_{11} = \underbrace{(-1)^{1+1}}_{2} \cdot d_{11}$

$C_{11} = 0$ $\quad C_{11} = d_{11}$

$d_{11} = \begin{vmatrix} 2 & 1 & 2 \\ 2 & 0 & 0 \\ -1 & 1 & 2 \end{vmatrix} \begin{matrix} 2 & 1 \\ 2 & 0 \\ -1 & 1 \end{matrix}$ $\quad d_{11} = 0$

$0 \quad 4 \quad 0 + 0 + 4$

$4 - 4 = 0$

$C_{31} = (-1)^{3+1} \cdot d_{31}$

$d_{31} = \begin{vmatrix} 4 & 2 & 4 \\ 2 & 1 & 2 \\ -1 & 1 & 2 \end{vmatrix} \begin{matrix} 4 & 2 \\ 2 & 1 \\ -1 & 1 \end{matrix}$

$-4 \quad 8 \quad 8 \quad 8 \quad -4 \quad 8$

$4 + 8 = 12 \quad 4 + 8 = 12$

$C_{31} = 0 \quad d_{31} = 12 - 12 = 0$

$C_{21} = \underbrace{(-1)^{2+1}}_{-1} \cdot d_{21}$

$C_{21} = -d_{21}$

$C_{21} = -0$

$d_{21} = \begin{vmatrix} 4 & 2 & 4 \\ 2 & 0 & 0 \\ -1 & 1 & 2 \end{vmatrix} \begin{matrix} 4 & 2 \\ 2 & 0 \\ -1 & 1 \end{matrix}$

$0 \quad 0 \quad 8 \quad 0 \quad 0 \quad 0$

$d_{21} = 8 - 8 = 0$

Resposta

$\cancel{1 \cdot 0}^{0} + \cancel{1 \cdot (-0)}^{0} \cdot \cancel{1 \cdot 0}^{0}$

$D(A) = 0$

4)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \end{bmatrix} ; \; B = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 2 \end{bmatrix}$$

$$(3A - 5B^t) = ?$$

primeiro calculamos a matriz transposta de B

Assim:

trocando linha por coluna ou vice-versa

$$B^t = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{bmatrix}$$

usando a propriedade de que um numero real multiplicado por uma matriz, multiplica cada elemento dela:

$$3 . A \rightarrow 3 . \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 6 & 9 \\ 6 & 3 & 12 \end{bmatrix}$$

$$5 . B^t \rightarrow 5 . \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 5 & 10 & 15 \\ 0 & 5 & 10 \end{bmatrix}$$

Subtraindo essas duas matrizes:

$$\begin{bmatrix} -2 & -4 & -6 \\ 6 & -2 & 2 \end{bmatrix}$$

3)

$$A = \begin{vmatrix} 3 & -2 \\ -4 & 3 \end{vmatrix} \qquad B = ?$$

$$B^2 = A$$

$$B \cdot B = A$$

$$B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}_{2\times 2}$$

$$\begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} \cdot \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -4 & 3 \end{bmatrix}$$

$$\begin{vmatrix} B_{11} \cdot B_{11} + B_{12} \cdot B_{21} & B_{11} \cdot B_{12} + B_{12} \cdot B_{22} \\ B_{21} \cdot B_{11} + B_{22} \cdot B_{21} & B_{21} \cdot B_{11} + B_{22} \cdot B_{22} \end{vmatrix}$$

$$\begin{vmatrix} (B_{11})^2 + B_{12} \cdot B_{21} & B_{11} \cdot B_{12} + B_{12} \cdot B_{22} \\ B_{21} \cdot B_{11} + B_{22} \cdot B_{21} & B_{21} \cdot B_{11} + (B_{22})^2 \end{vmatrix}$$

$$(B_{11})^2 + B_{12} \cdot B_{21} = 3$$

$$B_{11} \cdot B_{12} + B_{12} \cdot B_{22} = -2$$

$$B_{21} \cdot B_{11} + B_{22} \cdot B_{21} = -4$$

$$B_{21} \cdot B_{11} + (B_{22})^2 = 3$$